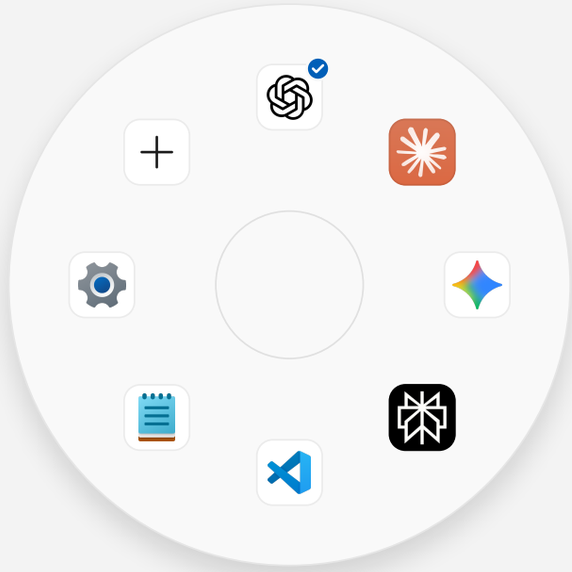
A quiet orbit on the Windows surface.

Copilot Reel opens one neutral plate at the pointer. Apps sit on a single orbit; the center stays silent until a name helps.



Light · Main, idle



Dark · Main, Claude selected

1 Windows-neutral surface

The system flyout surface, #F9F9F9 or #2C2C2C: one soft shadow, a 1 DIP rim, frosted when transparency is on.

2 One orbit

Every item sits on the 112 DIP orbit, slot 0 at 12 o'clock, clockwise, an equal step apart. Slots never reflow; icons never rotate.

3 A quiet center

Idle Main shows a faint ring and submenus a back chevron. Hover shows exactly one name, never a title, caption or hint.

4 Selection is a seat

A 48 DIP seat tinted with the system accent replaces the tile fill. Command glyphs and EN/ZH/JA stay foreground.

5 Marks stay true

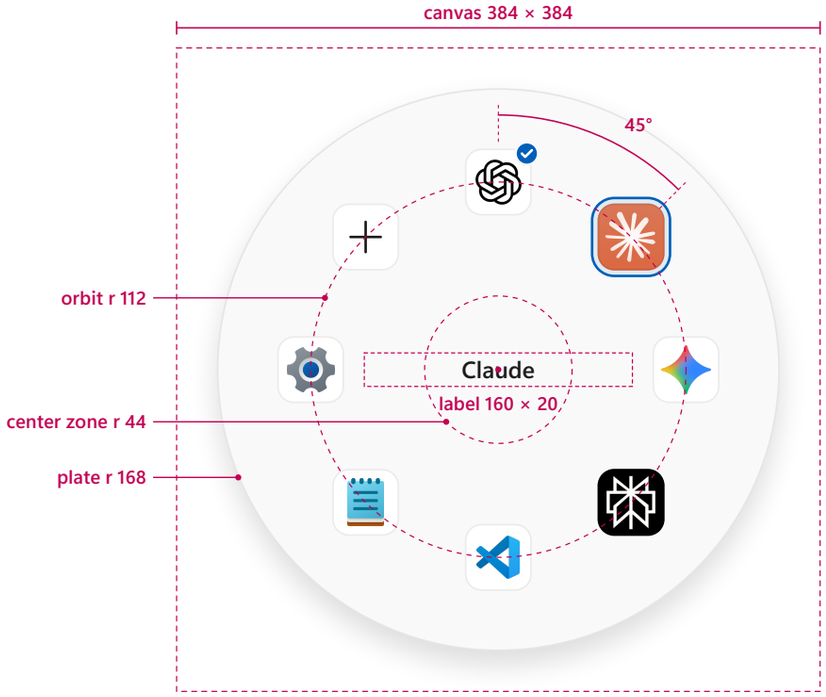
Vendor marks keep their own color and proportion. Plated marks fill the 40 DIP tile; open marks sit on a 28 DIP keyline.

6 Brief motion

83 to 250 ms, scaled from the plate center. Easing never overshoots; only the seat's spring does, by 0.6%. Reduced motion and high contrast settle at once.

Every dimension, in DIP.

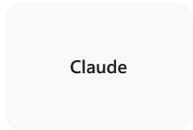
One canvas, one plate, one orbit. The selection seat is concentric with the tile; the center holds one line of text.



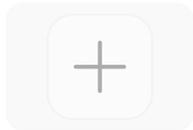
Slot 0 at 12 o'clock, clockwise; step 360/n (45° for n = 8).
Shadow: blur 16, offset y 8, inside the 24 DIP canvas margin.
Rim: 1 DIP, inset 0.5 (high contrast 2).
Pixel grid: centers, tiles and seats snap to device pixels; hairlines ≥ 1 px.

Idle Main
ring 1, ringIdle

Idle submenu
back chevron em 14

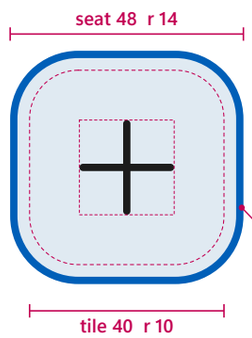


Named
one name, ring hidden



Disabled
tile and art at 0.36

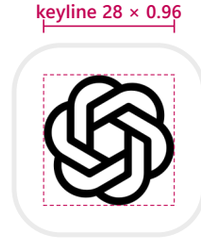
Tile and seat





High contrast: seat stroke 2.5. Vector fallback glyph stroke 1.75 (2).

Artwork



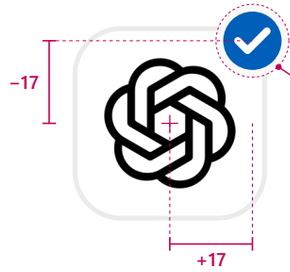
glyph
long ink side on the keyline



plated
hairline on top

Open marks sit on the keyline; plated marks fill the tile.

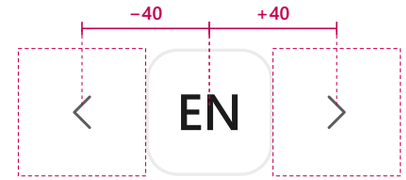
Preferred badge



disc r 6, accent
ring r 7.5: 1.5 DIP
plate-colored cutout
check 1.5, round caps,
textOnAccent

Accent disc, plate-colored cutout ring, path check.

Language tile



centers (±40, -112) from the plate center
hit 40 × 40 · em 14, ink 5.7 × 10.5
EN / ZH / JA 16 / 600, centered on cap height

EN / ZH / JA monogram; chevrons and A / D switch in place.

Roles, not colors.

Neutral surfaces from tokens, the accent from the Windows ramp, high contrast from live system colors.

	Light	Dark	High contrast COLOR_* system colors
plate	#F9F9F9	#2C2C2C	WINDOW
foreground	#1B1B1B	#FFFFFF	WINDOWTEXT
secondary	#5D5D5D	#CFCFCF	WINDOWTEXT
tile	#FFFFFF · frosted 70%	#FFFFFF · 10%	WINDOW, WINDOWTEXT 1 stroke
tileStroke	#000000 · 8%	#FFFFFF · 8%	WINDOWTEXT
plateStroke	#000000 · 8%	#FFFFFF · 10%	WINDOWTEXT, 2 DIP
ringIdle	#000000 · 10%	#FFFFFF · 10%	WINDOWTEXT
accent	AccentDark1 · #005FB8	AccentLight2 · #60CDFF	HIGHLIGHT
accentText	AccentDark2 · #003E92	AccentLight3 · #99EBFF	HIGHLIGHT
textOnAccent	#FFFFFF	#000000	HIGHLIGHTTEXT
selection	Mix(plate, accent, 0.1)	Mix(plate, accent, 0.18)	HIGHLIGHT, HIGHLIGHTTEXT 2.5 stroke
selectionText	foreground	foreground	HIGHLIGHTTEXT
outline	#CCCCCC · editor only	#4A4A4A · editor only	WINDOWTEXT
shadowOpacity	14%	26%	none

Accent is a role

Fill and text accents come from UISettings ramp steps, never the raw Accent. Fallbacks apply only when the ramp is unavailable.

The seat carries selection

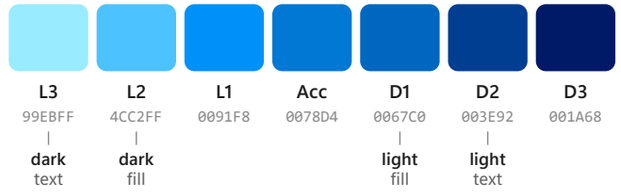
Command glyphs and EN/ZH/JA stay foreground on the seat. High contrast paints the seat COLOR_HIGHLIGHT and selected glyphs selectionText.

One mix rule

Mix(a, b, t) = a + (b - a) * t per channel, opaque. Translucent roles composite over the plate; frosted tiles use frostedTileAlpha.

System accent ramp

Sample: Windows default blue (capture machine)



Fallbacks when the ramp is unavailable: #005FB8 / #003E92 (light), #60CDFF / #99EBFF (dark). Scenes here use them.

Contrast (WCAG 2.2)

	Light	Dark	Need
Label, glyphs on plate	16.36	13.97	4.5:1
Glyphs on tile	17.22	10.19	4.5:1
Glyphs on seat	14.09	9.43	4.5:1
Chevrons on plate	6.25	8.96	3:1
Seat stroke on plate	5.99	7.76	3:1
Badge check on disc	6.31	11.67	3:1

High Contrast Black sample: text 21.0:1, selection 14.4:1.

In use



Light Dark High contrast

One name, centered on its cap height.

Segoe UI Variable Text 600 for Latin, Microsoft YaHei UI for Chinese, Reel Sans JP for Japanese. One line, never a second.

Segoe UI Variable Text · 600 · 14 DIP



Microsoft YaHei UI · Bold, the face DirectWrite picks for 600



Reel Sans JP · 600, bundled · page 16



Box 160 × 20 on the plate center. Baseline = center + capHeight × size / 2, capHeight 0.70 em from the Latin face, for every script.

EN ZH JA

EN ZH JA

The language is the icon

EN / ZH / JA names the interface language. Built-in labels are localized; user app names never are.

Truncation

One line in 160 DIP; character ellipsis trimming. Widths measured with the product face.

Visual Studio Code	119.7 DIP	fits
Windows Settings	114.1 DIP	fits
IntelliJ IDEA Ultimate	134.2 DIP	fits
DeepSeek Harness (DSH)	158.6 DIP	fits

Type tokens

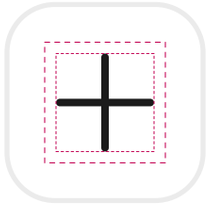
Label	14 DIP / 600, 1 line, 160 × 20, foreground
Monogram	EN / ZH / JA, 16 DIP / 600, cap-height centered, foreground
Latin	Segoe UI Variable Text (WWS); fallback Segoe UI
Chinese	Microsoft YaHei UI, zh-CN; then the system fallback
Japanese	Reel Sans JP 400 / 600, then Yu Gothic UI
CJK ranges	U+2E80-2FDF, U+3000-303F, U+3400-4DBF U+4E00-9FFF, U+F900-FAFF, U+FF00-FFEF
Kana ranges	U+3040-309F, U+30A0-30FF, U+31F0-31FF, U+FF65-FF9F
Locales	en-US default, zh-CN, ja-JP; kana make a name ja-JP
Never drawn	page titles, captions, helper copy A/D hints, Move here, a second label line

Four kinds of mark, one tile.

Command glyphs from Segoe Fluent Icons, open marks on a keyline, plated marks at full bleed, and one generic window.

Command

em 24 · ink 19.5



Segoe Fluent Icons at em 24, centered on the glyph's ink bounds, in foreground. Vector fallback strokes 1.75 DIP (high contrast 2), round caps and joins.

Glyph

keyline 28 × 0.96 = 26.9



An open mark on the neutral tile. Its long ink side fits the 28 DIP keyline × opticalScale. Also Notepad, Settings and custom apps via their Shell icons.

Plated

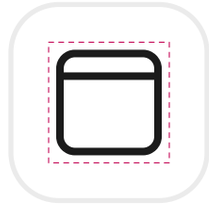
cover 40, clip r 10, hairline on top



A mark with its own filled plate: cropped to the plate, scaled to cover 40 DIP, clipped to r 10, hairline on top. No neutral tile beneath.

Generic

U+E737 at em 24



No artwork at all: window with a title bar (E737) on the neutral tile. Initials are never generated.

Command glyph map

Segoe Fluent Icons code points from tokens iconography.commandGlyphs; fallback Segoe MDL2 Assets, then vector paths. The badge check is always a path.

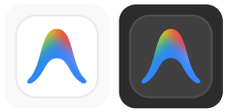
Add U+E710 Add · em 24	**Manage** U+F0E2 GridView · em 24	**Preferred** U+E930 Completed · em 24	**Sun** U+E706 Brightness · em 24	**Moon** U+E708 QuietHours · em 24	**Edit** U+E70F Edit · em 24	**Remove** U+E74D Delete · em 24
Previous U+E76B ChevronLeft · em 24	**Next** U+E76C ChevronRight · em 24	**ChevronLeftSmall** U+E76B ChevronLeft · em 14	**ChevronRightSmall** U+E76C ChevronRight · em 14	**BackChevron** U+E76B ChevronLeft · em 14	**GenericApp** U+E737 window · em 24	**TrayPreview** U+E890 View · em 24
TraySettings U+E713 Settings · em 24	**TrayDiagnostics** U+E9D9 Diagnostic · em 24	**TrayExit** U+E7E8 PowerButton · em 24	**About** U+E946 Info · em 24	**SwitchOn** U+EC11 ToggleFilled · em 24	**SwitchOff** U+F19E ToggleLeft · em 24	**CheckUpdate** U+E72C Refresh · em 24
CheckFailed U+EA6A SyncError · em 24	**Download** U+E896 Download · em 24					

Vendor marks, never redrawn.

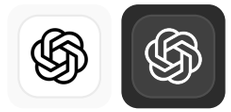
Each mark keeps its own color and proportion. The per-brand treatment is tokens iconography.brands, shown here at 1.25×.



JetBrains Air
`air`
plated · cover, clip r10



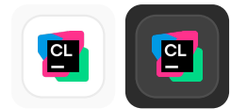
Antigravity
`antigravity`
glyph · 28.0 DIP (1)
`content 0.124, 0.156, 0.872, 0.844`



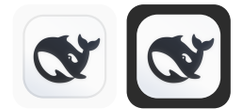
ChatGPT
`chatgpt-light / -dark`
glyph · 26.9 DIP (0.96)
`content 0.004, 0.008, 0.996, 0.992`



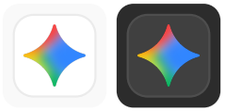
Claude
`claude`
plated · cover, clip r10



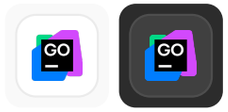
CLion
`clion`
glyph · 24.1 DIP (0.86)



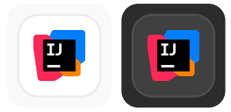
DeepSeek Harness
`dsh`
plated · cover, clip r10
`content 0.031, 0.031, 0.969, 0.969`



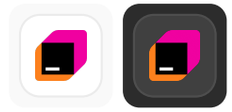
Gemini
`gemini`
glyph · 29.1 DIP (1.04)
`content 0.041, 0.041, 0.959, 0.959`



GoLand
`goland`
glyph · 24.1 DIP (0.86)



IntelliJ IDEA
`intellij-idea`
glyph · 24.1 DIP (0.86)



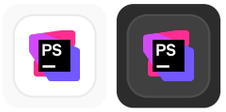
JetBrains
`jetbrains`
glyph · 24.1 DIP (0.86)



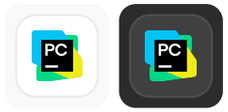
Junie
`junie`
glyph · 25.8 DIP (0.92)



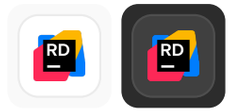
Perplexity
`perplexity`
plated · cover, clip r10



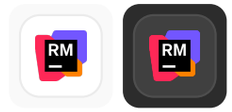
PhpStorm
`phpstorm`
glyph · 24.1 DIP (0.86)



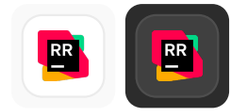
PyCharm
`pycharm`
glyph · 24.1 DIP (0.86)



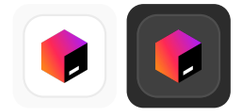
Rider
`rider`
glyph · 24.1 DIP (0.86)



RubyMine
`rubymine`
glyph · 24.1 DIP (0.86)



RustRover
`rustrover`
glyph · 24.1 DIP (0.86)



Toolbox
`toolbox`
glyph · 25.2 DIP (0.9)
`content 0.072, 0, 0.928, 1`



Visual Studio Code
`vscode`
glyph · 25.2 DIP (0.9)
`content 0, 0.002, 1, 0.998`



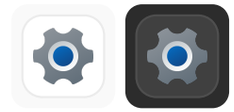
WebStorm
`webstorm`
glyph · 24.1 DIP (0.86)



ZCode
`zcode`
plated · cover, clip r10



Notepad
`notepad` · `Shell icon`
glyph · 28.0 DIP (1) · measured
`content 0.125, 0, 0.875, 1`



Windows Settings
`settings` · `Shell icon`
glyph · 28.0 DIP (1) · measured
`content 0, 0.027, 1, 0.969`

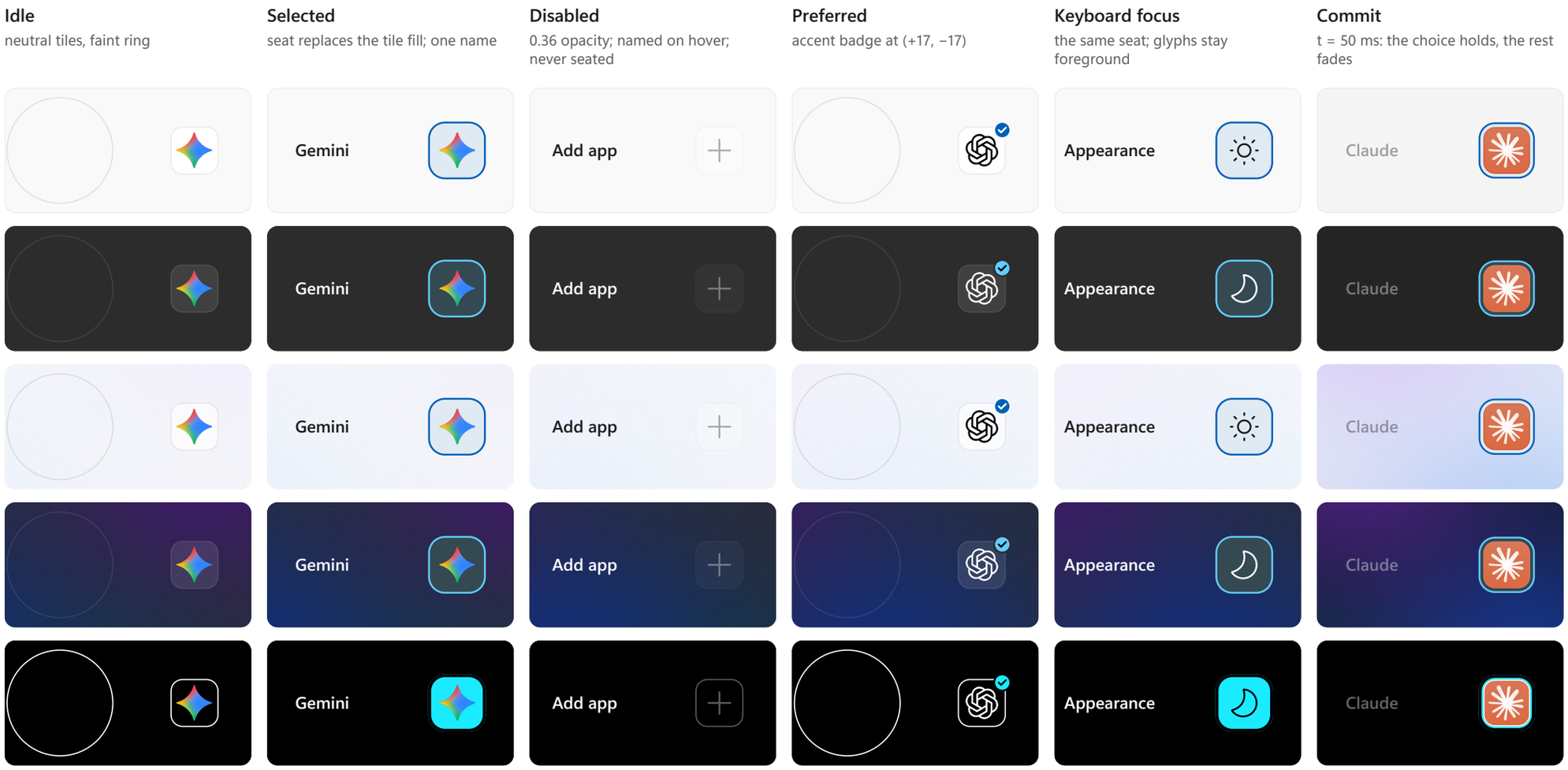


Custom app, no icon
`GenericApp glyph`
U+E737, em 24

Sources: the assets/brands PNGs the runtime decodes (SVG originals alongside); owner, source and hash in assets/brands/provenance.json. Content rects are normalized to the source canvas; "measured" = runtime alpha analysis.

Six states in every appearance.

Each cell is the slice from the plate center to the item at 3 o'clock. Selection is only ever the seat.



VECTOR Vector scenes from tokens.json; light and dark match the native captures (page 15). Frosted rows are tonal illustrations over a sample wallpaper. High contrast shows the High Contrast Black sample; the runtime paints with live GetSysColor values.

One orbit for two to nine items.

Slot 0 at 12 o'clock, then clockwise in equal steps of 360/n. At capacity the Add slot stays, disabled.



n = 2 · 180°
Manage apps, one app



n = 3 · 120°
App actions, last app



n = 4 · 90°
App actions



n = 5 · 72°
About



n = 6 · 60°
Main, five apps



n = 7 · 51.4°
The + hub



n = 7 · 51.4°
Presets, first page



n = 8 · 45°
Presets, middle page



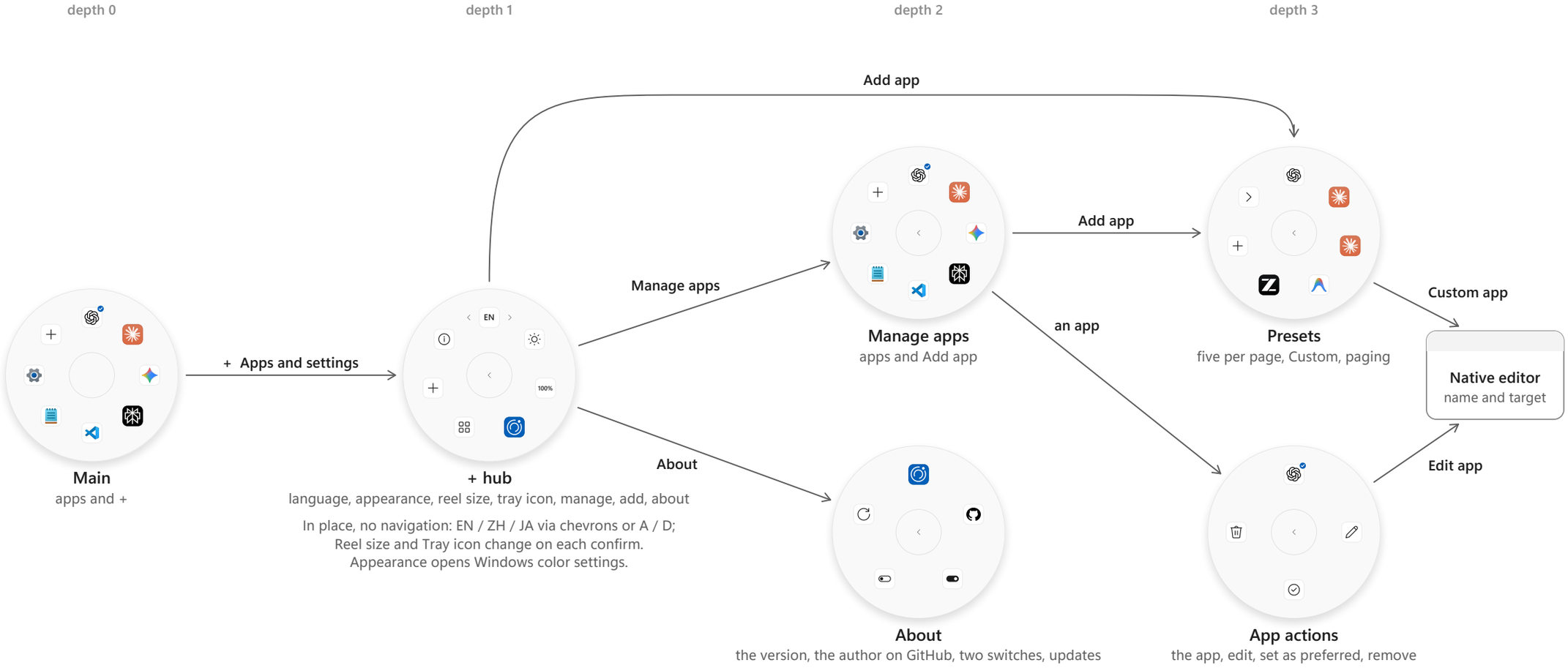
n = 9 · 40°
Main at capacity, 8 apps



n = 7 · 51.4°
Hub at 8 apps: Add disabled

Main, then one +.

Main holds the apps and a single +. Everything else hangs off the + hub; the center or Esc returns one level.



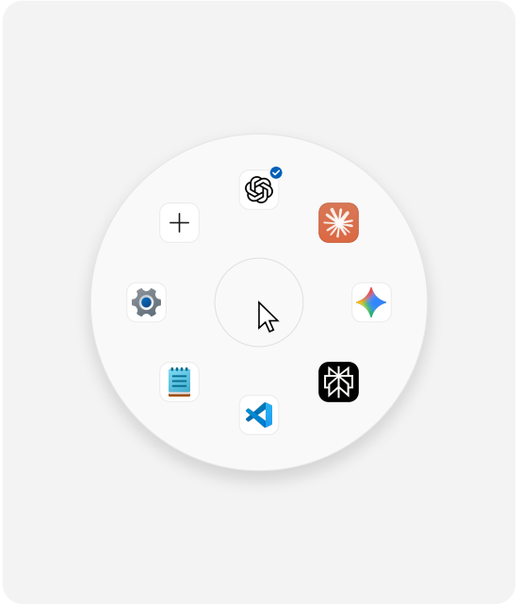
Deeper
DrillIn: the tapped tile travels to the center (200 ms) and becomes the back chevron.

Back
Center or Esc: DrillOut, one level; Main closes. Saving a choice also returns.

Paging
Previous / Next swap pages in place (120 ms).

Centered on the pointer; clamped only at edges.

The pointer is the plate center whenever the canvas fits. At an edge the reel moves inward just enough.

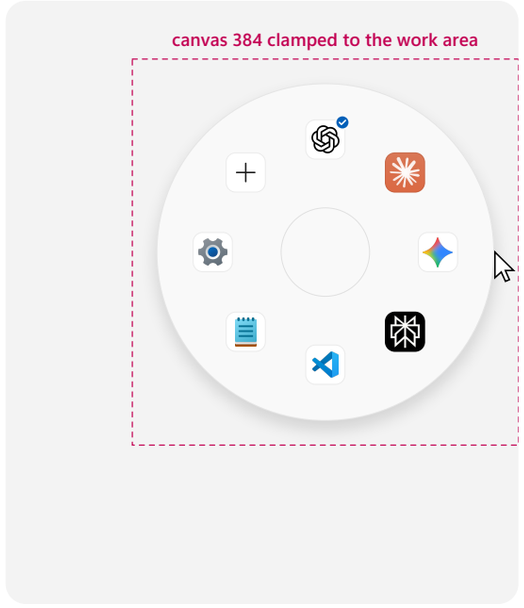


Interior

Pointer = plate center. Choose a direction at once.

Zero offset

Interior: pointer offset 0; no line from the pointer, no cursor warp.

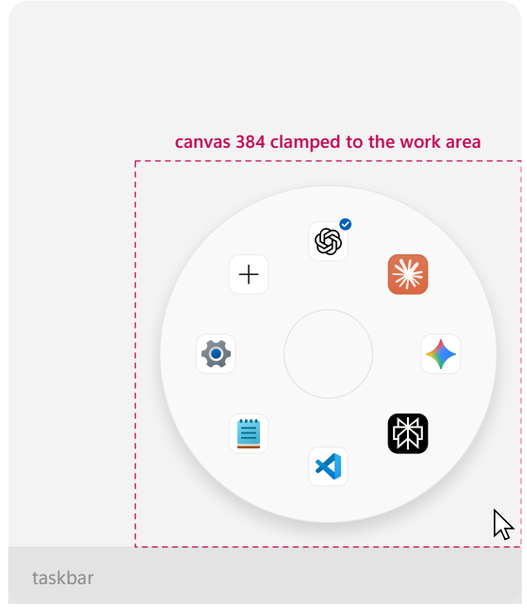


Screen edge

The 384 DIP footprint moves inward to the work area; the pointer may end up off the plate.

Minimal clamp

Clamp the whole canvas, shadow margin included (24 DIP); keep order, angles, radius and icon size.

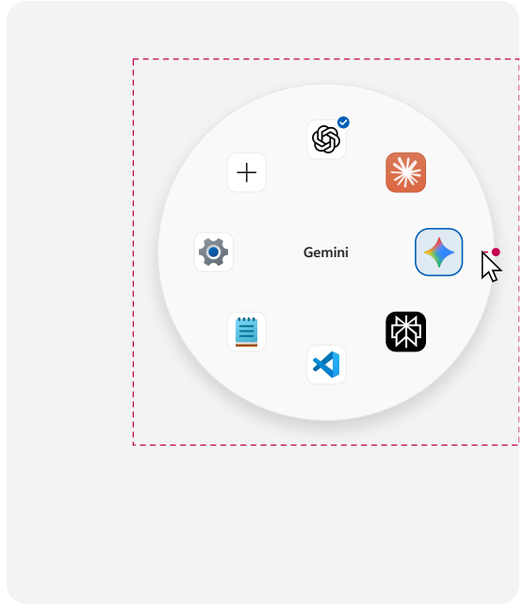


Corner

Both axes clamp minimally; the circle, order and angles stay.

No gate

On the plate or outside it (r 168), selection goes by direction once the pointer moves 4 DIP; the opening position never selects, and an early release cancels.



Off the plate

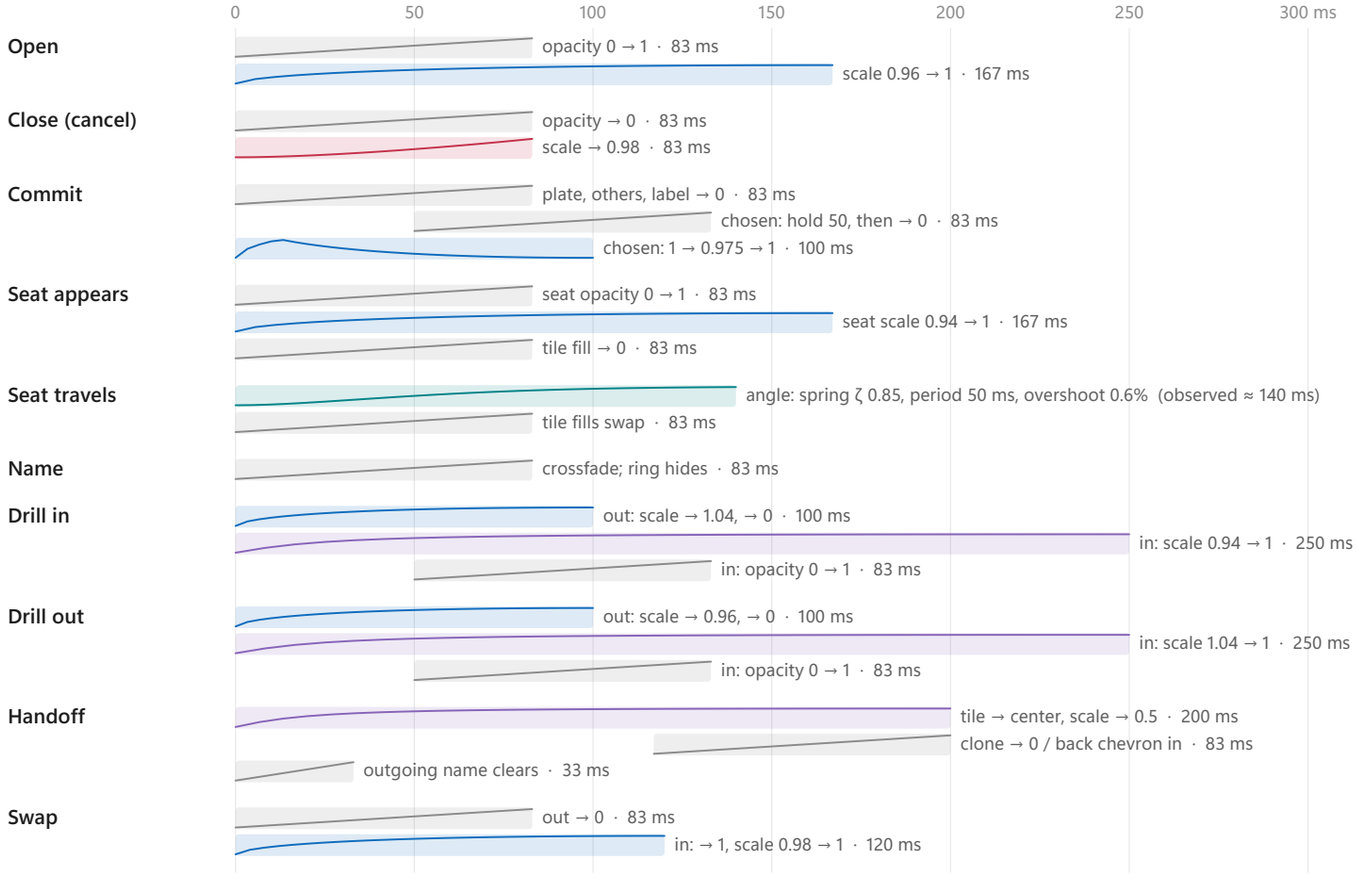
A pointer left outside the plate selects by direction as soon as it moves; it never enters the center first.

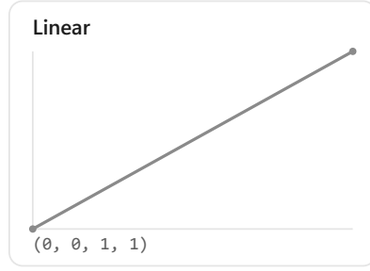
Small screens

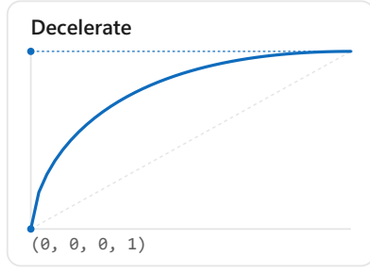
If the canvas cannot fit, scale it uniformly; text and targets shrink with it.

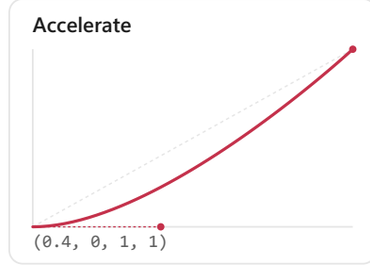
Brief, smooth, from the plate center.

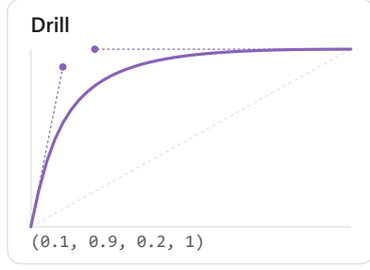
Full tier on Windows.UI.Composition: no app frame timers, easing without overshoot, every property from its current value.











Never

- Staggered item entrances
- Hover magnification or neighbor falloff
- Overshooting or elastic easing curves (the seat's spring aside)
- A rim direction notch
- Pointer-anchored transform origins
- Perimeter stroke animation

Origin: plate center (root CenterPoint); never the pointer.

Reduced motion and high contrast: the same code path with every duration 1 ms; completion still fires. Language EN / ZH / JA: only the monogram crossfades (83 ms).

The same motion, captured.

Frames from Windows Graphics Capture of the native 0.3.4 window (dark, solid), whose timings 0.3.13 keeps. Times are capture timestamps.

Open

open requested at 300 ms



Close

cancel at 51 ms



Seat

appears on Claude, then travels to Gemini



Drill in

+ clicked at 300 ms: handoff to the back chevron



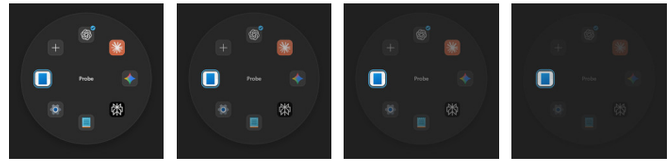
Drill out

Esc at 850 ms



Commit

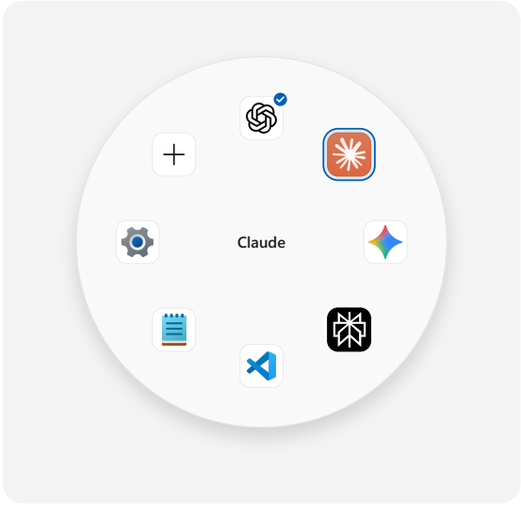
click at 300 ms



NATIVE output/runtime/v0.3.4/motion-open-select.png, motion-drill.png, motion-commit.png (frame strips at about 48 fps).
No later build recorded motion strips, so these frames show 0.3.4's five-item hub, drawn before the 0.8 reel scale.

Solid by default, frosted when Windows allows.

Transparency off: the opaque neutral plate. Transparency on: luminosity acrylic over the host backdrop, with the same tiles.



Light solid

VECTOR

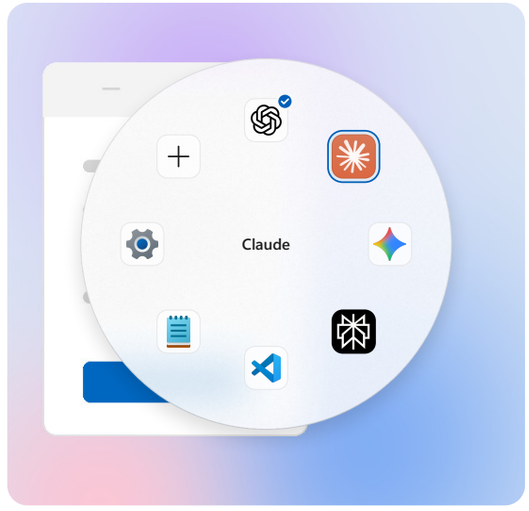
plate #F9F9F9, opaque
shadow 14%, blur 16, y +8
rim #000000 · 8%



Dark solid

VECTOR

plate #2C2C2C, opaque
shadow 26%, blur 16, y +8
rim #FFFFFF · 10%



Light frosted

TONAL

tint #FCFCFC, tint opacity 0
luminosity opacity 0.85
tile white at 0.7; fallback #F9F9F9



Dark frosted

TONAL

tint #2C2C2C, tint opacity 0.15
luminosity opacity 0.96
tile white at 0.1; fallback #2C2C2C

Acrylic recipe

Host backdrop	→	Luminosity blend	→	Color blend	→	Noise	→	Plate
blurred by the shell; no extra blur		tint RGB at luminosity opacity		tint at tint opacity		256 px tile, 2%, wrap, made once		transparent fill; rim and tiles on top

Solid whenever

- CompositionCapabilities reports effects unsupported or slow
- The backdrop cannot be created
- High contrast is on
- Energy saver is on

Rebuilt only when dark, transparency, high contrast or capabilities change.

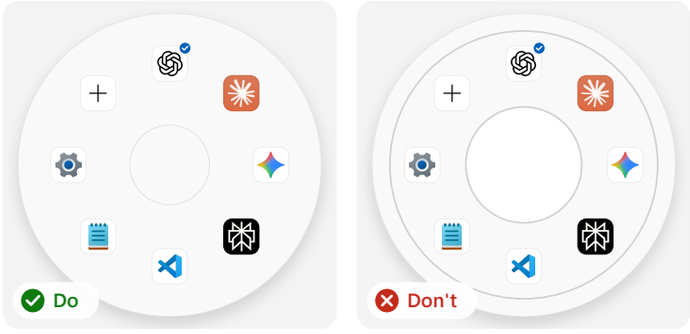
Layers, back to front

backdrop · shadow + plate + rim · ring · seat · page items · handoff · back + trash · labels ×2

Less, and only what the product draws.

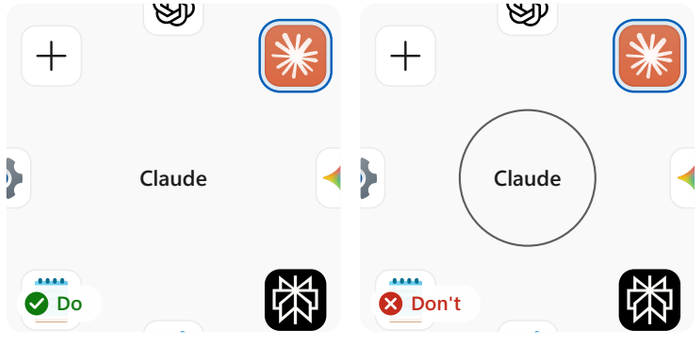
Six mistakes the native renderer cannot make, and the design it draws instead.

One plate



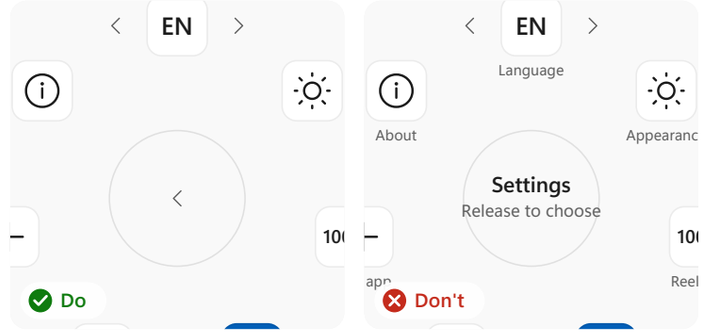
No plate inside the plate: no inner disc, panel or second rim.

The name owns the center



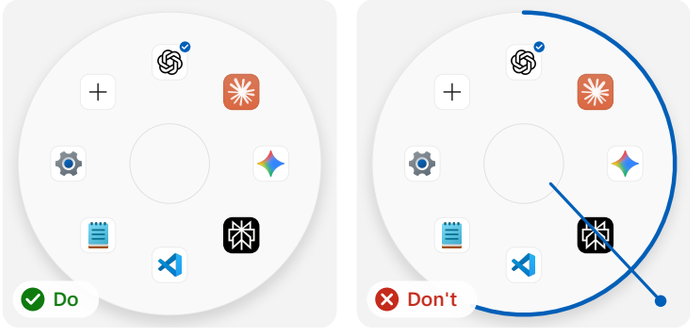
The ring hides while a name shows; never draw a label over the ring.

No captions, no helper copy



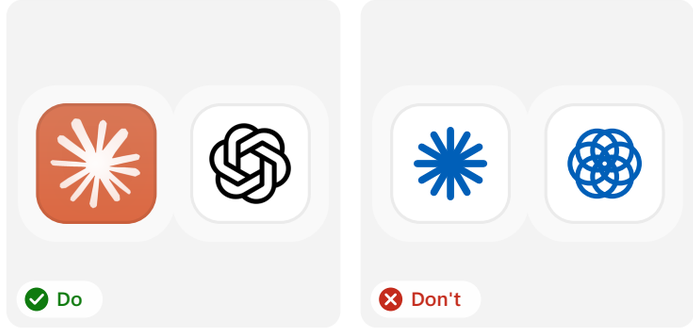
No titles, tile captions, hints or second lines. One name, on hover.

No sweep, no connector



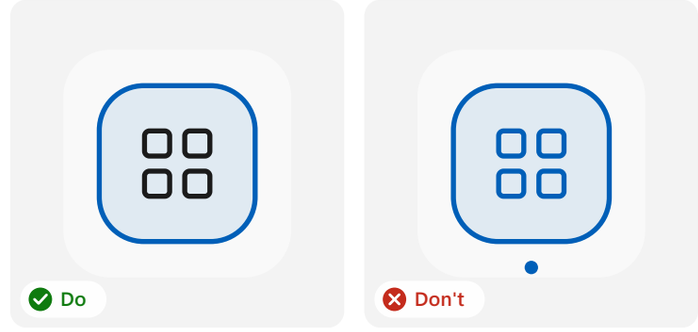
No accent perimeter stroke and no line from the pointer. The plate simply appears.

Marks stay true



Never recolor, outline or redraw a vendor mark (shown here schematically).

Glyphs stay foreground



The seat carries selection; no accent glyphs, no dot under the selection.

Verified in the native 0.3.13 build.

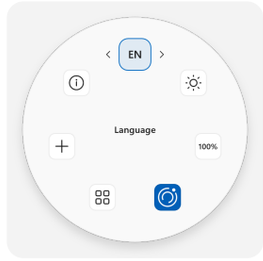
Windows Graphics Capture of the app window through the test seam, with the visual override; alpha background; 2x unless noted.



Light, Main



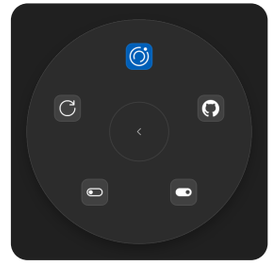
Light, Claude selected



Light, hub, Language



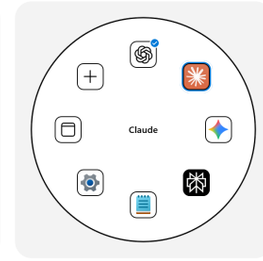
Dark, hub, Tray icon



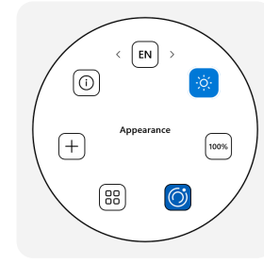
Dark, About



Dark, corner, by direction



Contrast path*, Claude



Contrast path*, Appearance

NATIVE

* Contrast path: the high-contrast code path with this machine's current system colors (not a High Contrast theme).

One tile at 100, 150 and 200 %

Dark, hub, Appearance selected; each crop enlarged to the same size, nearest neighbour.



100 % · 96 DPI



150 % · 144 DPI



200 % · 192 DPI

Evidence folder: output/runtime/v0.3.13, the reference set of tools/capture-seam.py (180 stills).
Motion strips: output/runtime/v0.3.4. Validation log: docs/validation.md.

Runtime-verified

– Light and dark solid plates: Main, the seven-item hub, About, Manage apps with drag and trash; tiles, seat, badge, switches, chevrons, name and ring

– The high-contrast code path (seat in COLOR_HIGHLIGHT) with current system colors

– Rendering at 100, 150 and 200 % through the DPI override

– Motion strips (0.3.4): open, cancel, seat appear and travel, drill in with handoff, drill out, commit

– EN / ZH (0.3.4): idle Main pixel-identical; the hub differs only inside the monogram

– Placement: zero interior offset, minimal clamp at eight edge and corner positions, no cursor warp; off the plate, selection by direction

Not verified yet

– The live frosted backdrop over real content (window capture omits the DWM host backdrop)

– Real High Contrast themes (Aquatic, Desert, Dusk, Night sky)

– Acrylic blend-mode mapping: tokens blendModeVerified is "pending"

– Physical Copilot-key timing (tap versus hold) on hardware

– Reduced motion, 144 / 240 Hz smoothness and GPU cost

Three scripts, one voice.

Apple's system pairing is the reference; the reel keeps Windows' own faces where they match it and bundles one where they do not.

Script	Apple reference	Windows system	Copilot Reel
Latin	SF Pro	Segoe UI Variable	**Segoe UI Variable Text 600** Manage apps
Chinese	PingFang SC	Microsoft YaHei UI	**Microsoft YaHei UI Bold** 管理应用
Japanese	Hiragino Sans	Yu Gothic UI	**Reel Sans JP 600** アプリの管理

Why Japanese is bundled

SF Pro, PingFang SC and Hiragino Sans share a large face, low stroke contrast, flat terminals and open counters, so mixed-script names sit evenly. Hiragino drops the uroko at stroke shoulders, centres each glyph's weight and draws kana slightly large. Fontworks' Rodin and UD faces share that build, but neither they nor Hiragino may ship inside another app without a licence, and Fontworks' open faces are a pencil script or geometric brand faces. Yu Gothic UI is hinted but draws condensed kana. Noto Sans JP (Source Han Sans) is the open face closest to Hiragino.

One fallback, chosen by locale

1. ja-JP text: Han, CJK punctuation and kana to Reel Sans JP, then Yu Gothic UI for what the subset lacks.
2. Other locales: kana to Reel Sans JP; Han and punctuation to Microsoft YaHei UI, which keeps mainland forms.
3. Everything else: the system fallback. A name that contains kana is laid out as ja-JP in every interface.

Reel Sans JP

– Noto Sans JP 2.004, static instances at 400 and 600

– Windows-31J, kana, CJK symbols and the interface strings: 7,617 characters, 2.4 MB a weight

– Segoe UI's line metrics, 1.079 / 0.251 em, so a Japanese line is as tall as an English one

– Renamed under the SIL Open Font License; OFL.txt ships beside it (tools/build-fonts.py)



NATIVE Hub at 96, 144, 192 DPI: Reel Sans JP (left), Yu Gothic UI (right)